The examination is being carried out on the **following application documents**

Description, Pages

1-31 filed with entry into the regional phase before the EPO

Claims, Numbers

1-10 filed with entry into the regional phase before the EPO

Drawings, Sheets

1/20-20/20 filed with entry into the regional phase before the EPO

1 Prior Art

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 SAKURAI RYO ET AL: "Emulating a sensor using soft material dynamics: A reservoir computing approach to pneumatic artificial muscle", 2020 3RD IEEE INTERNATIONAL CONFERENCE ON SOFT ROBOTICS (ROBOSOFT), IEEE, 15 May 2020 (2020-05-15), pages 710-717, XP033781288, DOI: 10.1109/ROBOSOFT48309.2020.9115974 [retrieved on 2020-06-12]

D2 ISHIKAWA TAKAHIRO ET AL: "Fault Analysis of Aging McKibben Pneumatic Artificial Muscle in Terms of its Model Parameters", 2018 IEEE CONFERENCE ON CONTROL TECHNOLOGY AND APPLICATIONS (CCTA), IEEE, 21 August 2018 (2018-08-21), pages 398-403, XP033430047, DOI: 10.1109/CCTA.2018.8511330 [retrieved on 2018-10-26]

D3 US 2003/065250 A1 (CHIEL HILLEL J [US] ET AL) 3 April 2003 (2003-04-03)

2 Article 54 EPC

2.1 The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claim 1 is not new within the meaning of Article 54(1) and (2) EPC in view of D1.

2.2 D1 discloses the subject-matter of claim 1 as follows (where references in parentheses refer to this document D1):

An estimation device comprising an estimation section that: (abstract "In this study, we propose a scheme that can estimate a PAM's length using applied pressure and its internal dynamics in real time, making use of the machine learning technique called reservoir computing. We show that estimation can be performed with significantly high accuracy.")

inputs a physical quantity indicating a plurality of member characteristics for an estimation target member (fig. 4A, case 3 Air pressure+resistance) into a learning model (fig. 4B) that:

employs, as training data, a plurality of sets of a physical quantity indicating a plurality of member characteristics of different types which change in time series according to deformation of a linearly- or nonlinearly-deforming member and a plurality of sets of a physical quantity indicating a performance state related to deformation of the member (section I, last paragraph "We investigated a method for estimating PAM length by sensing the dynamic electrical resistance of the soft material and utilizing it as time-series data for training the reservoir computing system."; fig. 6 "The red line shows the system output generalized using all the input conditions and the purple line shows the system output trained with the uniform input data condition of (Load, Step)=(210, 0.3). For each plot, the black line shows the target time series."), and

has been trained by being input with physical quantities indicating the plurality of member characteristics so as to output physical quantities indicating the performance state related to deformation of the member (fig. 4A; pg. 715, col. 2, par. 3 "The I/O scheme is fixed to Case 3, where we took air pressure and resistance as inputs and aim to output the corresponding length of the PAM."); and

estimates a physical quantity indicating a performance state related to deformation of the estimation target member.

(fig. 4A, fig. 5, fig. 6; pg. 715, col. 2, par. 3 "The I/O scheme is fixed to Case 3, where we took air pressure and resistance as inputs and aim to output the corresponding length of the PAM.")

2.3 As D1 discloses all features of claim 1, claim 1 lacks novelty (Article 54 EPC).

2.4 The same objections as for claim 1 apply mutatis mutandis to its corresponding independent claims 8 and 9 which are not novel either (Article 54 EPC).

2.5 The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claim 10 is not new within the meaning of Article 54(1) and (2) EPC in view of D1.

2.6 D1 discloses the subject-matter of claim 10 as follows (where references in parentheses refer to this document D1):

A learning model generation device comprising:

(pg. 713, col. 1, par. 5 "By training the ESN for each experimental condition, we evaluated its ability to estimate the PAM's length."; "pg. 713, col. 2, section B, par. 1 "The SR of the weights is adjusted according to the experiments comparing network performance to training data.")

an acquisition section that acquires a plurality of sets of:

a physical quantity indicating a plurality of member characteristics of different types which change in time series according to deformation of a linearly- or nonlinearly-deforming member, and

a physical quantity indicating a performance state related to deformation of the member; and

(section I, last paragraph "We investigated a method for estimating PAM length by sensing the dynamic electrical resistance of the soft material and utilizing it as time-series data for training the reservoir computing system."; fig. 6 "The red line shows the system output generalized using all the input conditions and the purple line shows the system output trained with the uniform input data condition of (Load, Step)=(210, 0.3). For each plot, the black line shows the target time series.")

a learning model generation section that, based on results of the acquisition by the acquisition section, generates a learning model (fig. 4B) that is input with physical quantities indicating the plurality of member characteristics (fig. 4A, case 3 Air pressure+resistance), and that is trained so as to output physical quantities indicating the

performance state related to deformation of the member. (fig. 4A; pg. 715, col. 2, par. 3 "The I/O scheme is fixed to Case 3, where we took air pressure and resistance as inputs and aim to output the corresponding length of the PAM.")

2.7 As D1 discloses all features of claim 10, claim 10 lacks novelty (Article 54 EPC).

2.8 Claim 5 lacks novelty (Article 54 EPC), since D1 discloses all additional features of this claim:

The estimation device of any one of claim 1 to claim 4, wherein the learning model is a model generated by training using a recurrent neural network. (pg. 713, col. 2, section B, par. 1 "The ESN is a recurrent neural network comprising internal computational nodes")

2.9 Claim 6 lacks novelty (Article 54 EPC), since D1 discloses all additional features of this claim:

The estimation device of any one of claim 1 to claim 5, wherein the learning model is a model generated by training using a network by reservoir computing.

(fig. 4B reservoir; section V, par. 3 "In this study, by using these inherent nonlinearities (calculated as the PAM's electrical resistance), we estimated the PAM length using the reservoir computing framework.")

3 Article 56 EPC

3.1 The subject-matter of dependent claim 2 does not involve an inventive step (Article 56 EPC), since D1 discloses the following non-crossed out additional features of this claim:

The estimation device of claim 1, wherein:
the member has an electrical characteristic that changes according to the deformation; (fig. 3 Resistance)
the physical quantity indicating the plurality of member characteristics includes a first physical quantity that indicates a pressure characteristic for deforming the member and a second physical quantity that indicates the electrical characteristic that changes according to the deformation of the member; (fig. 4A, case 3 Air pressure+resistance)
the physical quantity indicating the performance state related to deformation of the member includes a third

physical quantity that indicates a ~~number of repetitions of deformation~~ [length] of the member (fig. 4A Length of PAM); and the learning model is trained so as to output the third physical quantity with the first physical quantity and the second physical quantity as inputs. (fig. 4)

The differences identified above cannot support the presence of an inventive step since they are mathematical features which do not contribute with a specific technical effect to the solution of a technical problem because the calculation of data, i.e. estimates for the performance state of a deforming member, without its further use to actually manufacture such a member or perform indirect measurement for it, is not considered to address a technical problem (GL/EPO G-II, 3.3.2). The application appears to provide basis (fig. 5, 7, 15, [0036], [0038], [0085]) for amending the claim such that indirect measurement of the physical state of the system is performed by performing measurement of the input quantities (pressure and resistance) through technical means (various sensors as per [0036]). In such a hypothetical claim the differences identified above will make technical contribution to the prior art.

3.2 The subject-matter of dependent claim 3 does not involve an inventive step (Article 56 EPC), since D1 discloses the following non-crossed out additional features of this claim:

The estimation device of claim 2, wherein:
the member includes an elastic body that is formed with a hollow inside and that generates a contraction force in a specific direction due to a pressurized fluid being supplied to the hollow inside; (fig. 1A; pg. 712, section B, par. 1 "The PAM used in this study was composed of a dienic rubber tube reinforced with carbon black and encased in braided cords (see Fig. 1A)."; section II.A, par. 1 "The stress of a pressurized tube is transformed into an axial contraction force by means of braided cords whose geometry corresponds to a network of pantographs.")
the first physical quantity is a pressure characteristic indicating a plurality of pressure values in a time series in a case in which the pressurized fluid is supplied and supply of the pressurized fluid is cancelled (fig. 5A, input signals, pressure);
the second physical quantity is an electrical characteristic indicating a plurality of electrical resistance values in a time series of the elastic body,

which change according to the first physical quantity (fig. 5A, input signals, resistance); and
the third physical quantity is a performance indicator indicating a performance state (fig. 5A, output signal) ~~for a plurality of respective repetition number groups that result from dividing a predetermined specific number of repetitions, as a physical quantity indicating a performance state of the member while deformable and sustaining a specific performance, into a plurality of levels.~~

The differences identified above cannot support the presence of an inventive step since they are mathematical features which do not contribute with a specific technical effect to the solution of a technical problem because the calculation of data, i.e. estimates for the performance state of a deforming member, without its further use to actually manufacture such a member or perform indirect measurement for it, is not considered to address a technical problem (GL/EPO G-II, 3.3.2). The application appears to provide basis (fig. 5, 7, 15, [0036], [0038], [0085]) for amending the claim such that indirect measurement of the physical state of the system is performed by performing measurement of the input quantities (pressure and resistance) through technical means (various sensors as per [0036]). In such a hypothetical claim the differences identified above will make technical contribution to the prior art.

3.3 The subject-matter of dependent claim 4 does not involve an inventive step (Article 56 EPC), since D1 discloses the following non-crossed out additional features of this claim:

The estimation device of claim 2, wherein:
the member includes an elastic body that is formed with a hollow inside and that generates a contraction force in a specific direction due to a pressurized fluid being supplied to the hollow inside; (fig. 1A; pg. 712, section B, par. 1 "The PAM used in this study was composed of a dienic rubber tube reinforced with carbon black and encased in braided cords (see Fig. 1A)."; section II.A, par. 1 "The stress of a pressurized tube is transformed into an axial contraction force by means of braided cords whose geometry corresponds to a network of pantographs.")
the first physical quantity is a pressure characteristic indicating a plurality of pressure values in a time series in a case in which the pressurized fluid is supplied and

supply of the pressurized fluid is cancelled (fig. 5A, input signals, pressure);
the second physical quantity is an electrical characteristic indicating a plurality of electrical resistance values in a time series of the elastic body, which change according to the first physical quantity (fig. 5A, input signals, resistance); and
the third physical quantity is a ~~product lifespan~~ indicator that indicates a performance state (fig. 5A, output signal) ~~from a performance state at the number of repetitions of deformation of the member until a predetermined specific number of repetitions as a performance state of the member while deformable and sustaining a specific performance.~~

The differences identified above cannot support the presence of an inventive step since they are mathematical features which do not contribute with a specific technical effect to the solution of a technical problem because the calculation of data, i.e. estimates for the performance state of a deforming member, without its further use to actually manufacture such a member or perform indirect measurement for it, is not considered to address a technical problem (GL/EPO G-II, 3.3.2). The application appears to provide basis (fig. 5, 7, 15, [0036], [0038], [0085]) for amending the claim such that indirect measurement of the physical state of the system is performed by performing measurement of the input quantities (pressure and resistance) through technical means (various sensors as per [0036]). In such a hypothetical claim the differences identified above will make technical contribution to the prior art.

3.4 The subject-matter of dependent claim 7 does not involve an inventive step (Article 56 EPC), since D1 discloses the following non-crossed out additional features of this claim:

The estimation device of any one of claim 1 to claim 5, wherein the learning model is a model generated by training using a network by physical reservoir computing (fig. 4B reservoir; section V, par. 3 "In this study, by using these inherent nonlinearities (calculated as the PAM's electrical resistance), we estimated the PAM length using the reservoir computing framework."; section V, last paragraph "Lastly, this framework allows neuromorphic devices, such as spintronic devices [28][29] [30], to be implemented easily using physical reservoir computing.") employing a reservoir accumulated with ~~a plurality of sets of a physical quantity indicating an operation state of the~~

~~member,~~ a plurality of sets of a physical quantity indicating a member characteristic that changes according to the deformation, and a plurality of sets of a physical quantity indicating a performance of the member. (fig. 4B reservoir; The 2nd and 3rd sets coincide in D1)

The differences identified above cannot support the presence of an inventive step since they are mathematical features which do not contribute with a specific technical effect to the solution of a technical problem because the calculation of data, i.e. estimates for the performance state of a deforming member, without its further use to actually manufacture such a member or perform indirect measurement for it, is not considered to address a technical problem (GL/EPO G-II, 3.3.2). The application appears to provide basis (fig. 5, 7, 15, [0036], [0038], [0085]) for amending the claim such that indirect measurement of the physical state of the system is performed by performing measurement of the input quantities (pressure and resistance) through technical means (various sensors as per [0036]). In such a hypothetical claim the differences identified above will make technical contribution to the prior art.

3.5 D2 is also relevant to claims 1-10 for the inventive step objection, see pg. 398, col. 2, par. 2; pg. 399, col. 1, par. 1; pg. 400, col. 2, par. 1-4, pg. 401, section D, par. 1; pg. 401, section V, par. 1; table 1, 2. D3 is also relevant to claims 1-10 for the inventive step objection, see [0044], [0046], [0048]-[0051].

4 Article 84 EPC

4.1 Claims 1 and 10 are not concise (Article 84 EPC) as they exhibit undue repetition of claim wording. Claim 1 includes all features of claim 10. Claim 10 should thus be revised or deleted.

5 Closing remarks

5.1 It is not at present apparent which part of the application could serve as a basis for a new, allowable claim.

5.2 The applicant is reminded that:

a. the application may not be amended in such a way that it contains subject-matter which extends beyond the content of the application as filed (Article 123(2) EPC).

b. amendments made, irrespective of whether they concern amendments by addition, replacement or deletion, should be clearly identified (Rule 137(4) EPC).

c. the specific basis in the application as filed for each amendment should be indicated, on a feature-by-feature basis (Rule 137(4) EPC).

d. when filing amended claims the applicant should at the same time bring the description into conformity with the amended claims (Guidelines, F-IV, 4.3). In particular, this means that features required by the independent claims may not be presented in the description as being optional using wording such as "preferably", "may", "optionally", or only present in "some embodiments".

5.3 The applicant is reminded that documents D1-D3, which represent the closest prior art should be acknowledged in the description (Rule 42(1)(b) EPC).